Survey Supplement

No	Question	Data Type	Description / justification	Source
1	**Consent**	Binary categorical	Ethical prerequisite confirming voluntary participation and informed agreement.	
2-6	**Awareness questions**	Ordinal categorical	Establishes baseline awareness on AI use in financial services, opacity, problems in XAI and tradeoffs	
7	**Tradeoff preferences** On a scale from 0 to 10, which type of AI system in a financial app would you prefer?	Discrete numerical scale	Elicits the respondent's preference for interpretability versus predictive performance. This is asked by Habiba (2024) but we converted to a scale for public instead of Habiba's audience of ML Engineers	Habiba, Ue., Habib, M.K., Bogner, J. *et al.* How do ML practitioners perceive explainability? an interview study of practices and challenges. *Empir Software Eng* **30**, 18 (2025). https://doi.org/10.1007/s10664-024-10565-2
8-11	**General attitude toward artificial intelligence**	Ordinal Likert scale		Adapted from • Grassini, S. (2023). Development and validation of the AI attitude scale (AIAS-4): A brief

				measure of general attitude toward artificial intelligence. Frontiers in Psychology, 14, Article 1191628. https://doi.org/10.3389/fpsyg.2023.1191628 • Rohilla, T. (2024). *Effects of explainable artificial intelligence on trust in financial services digital platform ecosystems* (Doctoral dissertation, University of Reading). https://doi.org/10.48683/1926.00116750
12	Attention check	Categorical attention-check item	Data quality control	
	AI Finance Experience			
13	Have you used a money manager or investment app that used AI or smart automated tools?	Categorical screening item	Identifies whether respondents have direct experience with AI-enabled financial applications.	
14	Please provide the name or	Open-ended text	Provides contextual	Rohilla, T. (2024) *Effects of*

	description of the financial services platform that you are experienced or familiar with that likely uses AI.		information for interpreting experiences and identifying use cases.	*explainable artificial intelligence on trust in financial services digital platform ecosystems.* PhD thesis, University of Reading. doi: 10.48683/1926.00116750
15	How would you describe your interaction with the AI-powered financial app?	Multiple-response categorical	• Distinguishes participant roles, such as app user, business user, or developer. • Analyze interpretability with stakeholders of the ML system in mind. Simplified version adapted from Tomsett et al. (2018) • Participants can have more than one role (e.g., they use AI-powered financial tools for work)	Tomsett, Richard, Dave Braines, Dan Harborne, Alun Preece, and Supriyo Chakraborty. 2018. "Interpretable to Whom? A Role-based Model for Analyzing Interpretable Machine Learning Systems." arXiv. https://doi.org/10.48550/arXiv.1806.07552
16	Which activit(ies) did you engage in where the app likely made an automated decision (e.g., based on AI or algorithms)?	Multiple-response categorical	Identifies the specific financial activities in which respondents encountered automated decision-making.	

17	Other	Open-ended text		
18	**Branching questions** For any of the activities you selected, did the app provide you with an explanation?	Categorical branching item	Determines whether respondents actually received an explanation for an AI-supported financial decision.	
	Note on survey condition: ***1. Respondent received an explanation - Questions 19-24 and 28-40*** ***2. Respondent did not receive an explanation - Questions 25-27 and 28-40***			
	Explanation quality			
19	The explanation matched my level of knowledge and background.	Ordinal Likert scale; conditional	Measures whether the explanation was adapted to the respondent's expertise and comprehension level. This represents a user-adaptive explanation because the same technical information may need to be presented differently to different users.	Mohseni, S., Zarei, N., & Ragan, E. D. (2021). A multidisciplinary survey and framework for design and evaluation of explainable AI systems. *ACM Transactions on Interactive Intelligent Systems (TiiS)*, *11*(3-4), 1-45. Ehsan, U., Passi, S., Liao, Q. V., Chan, L., Lee, I. H., Muller, M., & Riedl, M. O. (2024, May). The

				who in XAI: how AI background shapes perceptions of AI explanations. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems* (pp. 1-32). Liao, Q. V., & Varshney, K. R. (2021). Human-centered explainable ai (xai): From algorithms to user experiences. *arXiv preprint arXiv:2110.10790*.
20	I understood how and why the recommendation was made.	Ordinal Likert scale; conditional	Measures perceived comprehensibility of the explanation. It captures whether the explanation helped the respondent form a meaningful understanding of the specific AI recommendation.	
21	The explanation gave me the right amount of detail—not too much, not too little.	Ordinal Likert scale; conditional	Measures explanation usability and perceived sufficiency. It recognizes that an explanation can be technically complete	Hoffman, R. R., Mueller, S. T., Klein, G., & Litman, J. (2018). Metrics for explainable AI: Challenges and

			but still unhelpful if it is too detailed or too vague.	prospects. *arXiv preprint arXiv:1812.04608*.
22	The explanation helped me judge whether or not to trust the decision.	Ordinal Likert scale; conditional	Measures whether the explanation supported appropriate trust and informed reliance. Explanations should help users evaluate a decision rather than simply encourage unconditional acceptance.	
22	How could the explanation have been more helpful or better suited to what you needed or wanted to know?	Open-ended text; conditional	Open-endedly identifies shortcomings in the explanation and allows respondents to describe desired improvements that may not be captured by fixed-response options.	
23	If the app had provided the improvements you mentioned, how much would your trust in its decision have changed?	Discrete numerical scale; conditional		Studies relating XAI to trust or trust calibration with a financial system 1. Dikmen, M., & Burns, C. (2022). The effects of domain knowledge on trust in explainable AI and task performance: A case of peer-to-peer lending. *International*

				Journal of Human-Computer Studies, *162*, 102792. 2. Han, J., & Ko, D. (2025). Trust formation, error impact, and repair in human–AI financial advisory: A dynamic behavioral analysis. *Behavioral Sciences*, *15*(10), 1370.
	Experience without explanation *These questions are qualified for respondents who answered "No—I did not receive an explanation" in Question 18.*			
25	When you use the app, how did you feel about the outcome, recommendation, or decision, even without an explanation?	Ordinal categorical; conditional	Measures respondents' trust or emotional response when an AI decision is provided without an explanation.	
26	What type of explanation were you hoping to see?	Open-ended text; conditional	Identifies the explanation content or format that respondents wanted but did not receive.	
27	If the app had provided an explanation that matches your preferences, how much would your trust in its decision have changed?	Discrete numerical scale; conditional	Measures the expected effect of receiving a preferred explanation after an unexplained decision.	See 23.
	Explanation Preferences			

	These questions are intended for both respondents who received an explanation and those who did not.			
28	In which of the following cases do you think an AI system should be required to provide an explanation?	Multiple-response categorical	• EU AI Act Chapter III has classifications for which tasks require explanations • Example of non-high risk include using AI to calculate capital requirements of credit and insurance institutions for prudential purposes or detecting financial fraud, but whether this is seen by public as needed is what is asked by this question	
29	A clear explanation of how the app decided your result (e.g., how your spending or credit history affected your score)	Ordinal importance scale		Questions 29-36;38-40 were taken from a mixture of the ff sources: 1. Transforming classical XAI like feature importance to descriptive for the purpose of the survey 2. 12 conceptual properties developed by Nauta, M., Trienes, J., Pathak, S., Nguyen, E., Peters, M., Schmitt, Y., ... & Seifert, C. (2023). From anecdotal evidence to quantitative evaluation methods: A systematic review on evaluating explainable ai. *ACM Computing Surveys*, *55*(13s), 1-42.
30	Knowing the main factors that influenced your result (e.g., income, spending habits, payment history)	Ordinal importance scale	Feature-based explanation like SHAP or LIME	
31	Seeing how your result compares with expert opinions or standard benchmarks (e.g., market averages or official credit criteria)	Ordinal importance scale		
32	Seeing how confident or reliable the app's prediction is (e.g., how certain it is that a transaction is suspicious)	Ordinal importance scale		
33	Understanding which parts of the process were automated and which involved a person	Ordinal importance scale	Human-in-the-Loop	

34	Learning how errors are detected and corrected by the app (e.g., when incorrect information affects your account)	Ordinal importance scale		https://arxiv.org/abs/2201.08164 3. Lopes, P., Silva, E., Braga, C., Oliveira, T. B. de, & Rosado, L. (2022). XAI systems evaluation: A review of human- and computer-centred methods. *Applied Sciences, 12*(19), Article 9423. https://doi.org/10.3390/0 4. Kuiper, O., van den Berg, M., van der Burgt, J., & Leijnen, S. (2021, November). Exploring explainable ai in the financial sector: Perspectives of banks and supervisory authorities. In *Benelux Conference on Artificial Intelligence* (pp. 105-119). Cham: Springer International Publishing. a. This paper establish a list of explanation types beyond the model including
35	Knowing whether a human or an AI made or reviewed the decision (e.g., loan approval, fraud alert)	Ordinal importance scale		
36	Being informed about possible risks or consequences if the AI system makes an error (e.g., a transaction being blocked by mistake)	Ordinal importance scale		
37	This is an attention check. Please select “Don’t Know.”	Categorical attention-check item	Attention Check	
38	Getting advice on how to improve or change the outcome (e.g., how to increase your credit score)	Ordinal importance scale	Counterfactual explanation or algorithmic recourse	
39	Getting a personalized explanation for your specific case (e.g., why a certain investment was recommended over another)	Ordinal importance scale		
40	Knowing how your data is used and protected in the AI system	Ordinal importance scale		

				process-based explanations
	Responsibility and governance			
41	Who should be responsible for ensuring AI systems in finance are transparent?	Single-response categorical	Identifies which actors respondents believe should be accountable for transparency.	Tóth, Z., & Blut, M. (2024). Ethical compass: The need for Corporate Digital Responsibility in the use of Artificial Intelligence in financial services. *Organizational Dynamics*, *53*(2), 101041.
42	Other governing body	Open-ended text; conditional	Allows respondents to specify an accountability actor not included in Question 41.	
	Demographic and occupational information			
43	Gender	Categorical	Collects demographic information for subgroup analysis.	Many studies relate background of explained to the type of explanation. 1. Country a. West vs East - Chien, S. Y., Wang, Y. F., Cheng, K. T., & Chen, Y. C. (2025). Comparative study of XAI perception between eastern and
44	Age	Ordinal categorical	Enables analysis of differences across age groups.	
45	Country	Categorical or open-ended country field	Enables comparison across national, cultural, and regulatory contexts.	
46	Which one best describes your job level?	Categorical	Measures professional seniority and work position.	
47	Others (job title)	Open-ended text;	Allows respondents to	

		conditional	provide a job title not represented in the predefined categories.	western cultures. *International Journal of Human–Computer Interaction*, *41*(17), 11192-11208. 2. Gender and Age / Individual Characteristics a. Reeder, S., Jensen, J., & Ball, R. (2023, July). Evaluating explainable AI (XAI) in terms of user gender and educational background. In *International Conference on Human-Computer Interaction* (pp. 286-304). Cham: Springer Nature Switzerland. b. Wienrich, C., Carolus, A.,
48	For students: What is your field of study?	Categorical	Captures respondents' academic background and potential financial or technological literacy.	
49	Which sector do you belong to?	Categorical	Identifies the respondent's occupational sector and possible exposure to financial technologies.	

				Roth-Isigkeit, D., & Hotho, A. (2022). Inhibitors and enablers to explainable AI success: a systematic examination of explanation complexity and individual characteristics . *Multimodal Technologies and Interaction*, *6*(12), 106. 3. Cognitive style and sociocultural factors (gender, religion, educaiton) - Kopecka, H., Such, J., & Luck, M. (2024). Preferences for AI explanations based on cognitive style and socio-cultural factors. *Proceedings of the ACM on Human-Computer Interaction*, *8*(CSCW1), 1-32.